avail; they knew so much more about quantum fields in curved spacetime than I that although (as usual) I was quite sure I had truth on my side, I could not counter their arguments.

My return flight to America was scheduled for Tuesday, 23 September. On Monday evening, as I was packing my bags in my tiny room at the University Hotel, the telephone rang. It was Zel'dovich: "Come to my flat, Kip! I want to talk about black-hole evaporation!" Tight for time, I sought a taxi in front of the hotel. None was in sight, so in standard Muscovite fashion I flagged down a passing motorist and offered him five rubles to take me to Number 2B Vorobyevskoye Shosse. He nodded agreement and we were off, down back streets I had never traveled. My fear of being lost abated when we swung onto Vorobyevskoye Shosse. With a grateful "Spasibo!" I alighted in front of 2B, jogged through the gate and forested grounds, into the building, and up the stairs to the second floor, southwest corner.

Zel'dovich and Starobinsky greeted me at the door, grins on their faces and their hands above their heads. "We give up; Hawking is right; we were wrong!" For the next hour they described to me how their version of the laws of quantum fields in a black hole's curved spacetime, while seemingly different from Hawking's, was really completely equivalent. They had concluded black holes cannot evaporate because of an error in their calculations, not because of wrong laws. With the error corrected, they now agreed. There is no escape. The laws require that black holes evaporate.

There are several different ways to picture black-hole evaporation, corresponding to the several different ways to formulate the laws of quantum fields in a black hole's curved spacetime. However, all the ways acknowledge vacuum fluctuations as the ultimate source of the outflowing radiation. Perhaps the simplest pictorial description is one based on particles rather than waves:

Vacuum fluctuations, like "real," positive-energy waves, are subject to the laws of wave/particle duality (Box 4.1); that is, they have both wave aspects and particle aspects. The wave aspects we have met already (Box 12.4): The waves fluctuate randomly and unpredictably, with positive energy momentarily here, negative energy momentarily there, and zero energy on average. The particle aspect is embodied in the concept of *virtual particles,* that is, particles that flash into existence in pairs (two particles at